

CAP470: CLOUD PLATFORM AND SERVICE MODEL

L:3 T:0 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: associate the basic concepts of cloud computing, architecture and its applications

CO2 :: examine different types of virtualisation techniques.

CO3 :: illustrate the cloud storage file systems and various security standards

CO4 :: evaluate the technical challenges in design of collaborative cloud applications, Authentication and Authorization

Unit I

Introduction to cloud computing : cloud overview, definition of cloud computing, evolution of cloud computing, components of cloud computing, characteristics of cloud computing, issues of cloud computing, advantage and disadvantages, applications of cloud computing, cloud platforms, Comparison of existing Cloud platforms /Web Services, Cloud computing optimization

Unit II

Cloud computing architecture : service models: infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS), deployment models: public cloud, private cloud, hybrid cloud, community cloud, business models: NIST cloud computing reference model. cloud cube model, Cloud Infrastructure Performance, Networking in cloud, Resource Management

Unit III

Cloud virtualization : virtualization, components of virtualization, characteristics of virtualization, types of virtualization, advantage and disadvantages of virtualization, virtual machine, properties of virtual machines, hypervisor, types of hypervisor, containerization, benefits of containerization, container architecture elements

Containers : Introduction, core concepts, docker, Container Orchestration, Use Cases, Advantages and Security

Unit IV

Cloud access, cloud storage and file systems : platforms, web applications, web API's, web browser, storage system in cloud, types of storage systems, big data, MapReduce, hadoop framework, hadoop distributed file system (HDFS), google file system (GFS), storage volume, scalability

Unit V

Cloud security and standards : cloud security challenges, authorization, authentication, identify & access management, data security, data integrity, encryption & key management, open cloud consortium (OCC), distributed management task force (DMTF), standards for application developers, standards for messaging, standards for security

Unit VI

Cloud application design and collaboration : cloud application design considerations, cloud application reference architectures, design methodologies, data storage, data analytics, deployment & management, calendars, schedules and task management, project management, web- based communication tools, Introduction to Green Cloud

Authentication and Authorization – AWS (IAM) : Introduction, IAM Identities, IAM Policies, IAM Users, IAM Groups, IAM Roles, Authentication Tools

Text Books:

1. CLOUD COMPUTING, A PRACTICAL APPROACH by TOBY VELTE, ANTHONY VELTE, ROBERT ELSENPETER, Mc Graw Hill Education

References:

1. CLOUD COMPUTING: CONCEPTS, TECHNOLOGY & ARCHITECTURE by THOMAS ERL, Pearson Education India
2. CLOUD COMPUTING BIBLE by BARRIE SOSINSKY, WILEY

